Corporate Finance - Problem Set 2

Exercise 1 Consider an economy with complete markets and two equally probable states: slump and boom. The market prices and payoffs of a stock and a bond are given by:

		Payoff (t=1)	
	Price (t=0)	Slump	Boom
Stock	1.05	0	2.5
Bond	0.92	1	1

A penniless entrepreneur has access to the following project:

	Payoff (t=1)	
Investment (t=0)	Slump	Boom
1	0.5	3

The assumptions of the Modigliani-Miller framework are satisfied, including zero corporate taxes.

(a) What is the no-arbitrage price of a contingent claim that pays 1 in the slump state and zero otherwise? What is the no-arbitrage price of a contingent claim that pays 1 in the boom state and zero otherwise?

(b) Denote by $C(K)$ the no-arbitrage price of a call option on the project assets with strike K. Find $C(1)$ and $C(1.5)$.

(c) Denote by $P(K)$ the no-arbitrage price of a put option on the project assets with strike K. Find $P(1)$ and $P(1.5)$.

(d) Show that

$$C(1) + \frac{1}{(1/0.92)} - P(1) = C(1.5) + \frac{1.5}{(1/0.92)} - P(1.5)$$

Interpret the result.

(e) Suppose the entrepreneur decides to raise 1 by issuing debt. What would be the face value of debt? Find r_D, r_E, r_A, E, A and D.

(e) Suppose the entrepreneur decides to raise an amount D by issuing debt, and the remainder $1 - D$ by issuing equity. What would be the face value of debt for a given D? Find r_D, r_E, r_A, E, A and D for all $D \in \{0.01, 0.02, 0.03, ..., 0.98, 0.99, 1\}$.

(f) The penniless entrepreneur needs to consume a minimum amount at $t = 0$. Suppose the entrepreneur decides to raise 1.1 by issuing debt, out of which 0.1 will be used to pay a dividend at $t = 0$, and the remainder 1 will be used to finance the project. What is the value of the firm after the dividend is paid? Is the entrepreneur better off?

Exercise 2 Company A currently has no debt in its capital structure and 10 million shares outstanding. The beta of its equity is 1.20. Company A's expected future free cash flow is \$25 million per year in perpetuity, starting next year. Company A is considering changing its capital structure by issuing debt and using the proceeds to buy back stock in such a way that it will have a constant debt-equity ratio of 25% after the change (continuously adjusted). Assume that A's debt cost of capital will be 3%. The corporate tax rate is 25%, and there are no other market imperfections. Assume that the CAPM holds, the risk-free rate is 3%, and the expected return on the market is 8%.

(a) Compute the rate of return of the equity before and after the change in the capital structure.

(b) Compute the firm's new WACC.

(c) Compute the value of the all-equity firm and the stock price before and after the announcement of the change. Find the value of the levered firm and the tax shield (after the change). Find the stock price and the new number of outstanding shares after the issue.

(d) Assume only for this question that company A issues debt, but instead of keeping a constant debt-equity ratio equal to 25% the firm keeps interest expenses equal to 20% of its free cash flow. Compute the value of the tax shields in this new situation.

(e) Assume only for this question that company A issues debt, but instead of keeping a constant debt-equity ratio equal to 25% the firm issues \$100 million of perpetual debt. Compute the value of the firm's tax shields, the stock price, and the new number of outstanding shares after the issue.

Exercise 3 Consider an entrepreneur with a project that requires an investment of $I = 60$ today. The project generates (unlevered) free cash flow of 100 or 60 tomorrow with probability 50% each. The expected return on assets is $r_A = 15\%$. Assume there are no market imperfections and corporate taxes are zero.

(a) Find the value of the all-equity firm.

(b) Suppose the entrepreneur chooses to raise 60 by issuing debt and that creditors demand a return of 6.6%. What is the face value of debt? What is the market value of equity?

(c) Now assume that if bankruptcy occurs, the firm must pay 7% of the realized cash flow to lawyers (direct bankruptcy costs). Find the face value of debt and the market value of equity. Which party is "paying" for the bankruptcy costs?

Exercise 4 Consider the Modigliani-Miller framework. Yerba Industries is an all-equity firm whose stock has a beta of 1.2 and an expected return of 12.5%. Suppose it issues new risk-free debt with a 5% yield and repurchases 40% of its stock. Assume perfect capital markets.

(a) What is the beta of Yerba stock after this transaction?

(b) What is the expected return of Yerba stock after this transaction?

Suppose that before this transaction, Yerba expected earnings per share this coming year

of £1.50, with a forward P/E ratio (that is, the share price divided by the expected earnings for the coming year) of 14.

(c) What is Yerba's expected earnings per share after this transaction? Does this change benefit shareholders? Explain.

(d) What is Yerba's forward P/E ratio after this transaction? Is this change in P/E ratio reasonable? Explain.